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VEHICLE FAULT PREDICTION SYSTEM v2.0
Two-Level Diagnostic Architecture
Based on SAE J1979 Standard
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[1/4] Generating training dataset from SAE J1979 thresholds...
    Generated 4800 training samples (7 fault classes + normal)
[2/4] Training neural network (MLP 12->32->16->8)...
    Training complete. Best epoch: 92
    Test Accuracy: 82.29%

[3/4] System ready.
[4/4] Starting interactive interface...

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MAIN MENU
-----
1. Manual diagnosis (enter sensor values yourself)
2. Random simulation (system randomizes parameters)
3. Exit
-----

Select option (1/2/3): 1

Enter current OBD-II sensor readings:
(Press ENTER to use default normal value)

Engine Coolant Temperature (C) [normal: 92.0]: 1
Engine RPM (rpm) [normal: 1500.0]: 1
Engine Load (%) [normal: 35.0]: 1
Intake Air Temperature (C) [normal: 25.0]: 1
Short Term Fuel Trim (%) [normal: 0.0]: 1
Fuel Tank Pressure (kPa) [normal: 0.0]: 1
Mass Air Flow Rate (g/s) [normal: 8.0]: 1
Throttle Position (%) [normal: 20.0]: 1
Battery Voltage (V) [normal: 14.0]: 1
O2 Sensor Voltage (V) [normal: 0.5]: 1
Vehicle Speed (km/h) [normal: 60.0]: 11
Component Wear Index (min) [normal: 50.0]: 111

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DIAGNOSIS REPORT – MANUAL INPUT
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LEVEL 1: Parameter Threshold Analysis (SAE J1979)
-----
[HIGH [RED]] Battery Voltage
    Value: 1.00 V << RED threshold (11.5)

LEVEL 2: Neural Network Fault Prediction (MLP)
-----
Top predicted faults:
1. Electrical / Charging Fault          65.1%
2. Transmission / Speed Anomaly        33.0%
3. Fuel System Malfunction             1.3%

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FINAL DIAGNOSIS
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STATUS : ANOMALY DETECTED (fault type unclear)
ALERT : HIGH [RED]
CONFIDENCE : 65.1% (below threshold for fault ID)
ACTION : Inspect anomalous parameters. Seek professional diagnosis.

ANOMALOUS PARAMETERS:

- Battery Voltage: 1.00 V

MAIN MENU

1. Manual diagnosis (enter sensor values yourself)
2. Random simulation (system randomizes parameters)
3. Exit

Select option (1/2/3): 1

Enter current OBD-II sensor readings:
(Press ENTER to use default normal value)

Engine Coolant Temperature (C) [normal: 92.0]: 20000
Engine RPM (rpm) [normal: 1500.0]:
Engine Load (%) [normal: 35.0]:
Intake Air Temperature (C) [normal: 25.0]:
Short Term Fuel Trim (%) [normal: 0.0]:
Fuel Tank Pressure (kPa) [normal: 0.0]:
Mass Air Flow Rate (g/s) [normal: 8.0]:
Throttle Position (%) [normal: 20.0]:
Battery Voltage (V) [normal: 14.0]:
O2 Sensor Voltage (V) [normal: 0.5]:
Vehicle Speed (km/h) [normal: 60.0]:
Component Wear Index (min) [normal: 50.0]:

DIAGNOSIS REPORT – MANUAL INPUT

LEVEL 1: Parameter Threshold Analysis (SAE J1979)

[HIGH [RED]] Engine Coolant Temperature
Value: 20000.00 C >> RED threshold (120.0)

LEVEL 2: Neural Network Fault Prediction (MLP)

Top predicted faults:

- | | |
|------------------------------|-------|
| 1. Fuel System Malfunction | 96.3% |
| 2. Exhaust / Emissions Fault | 3.3% |
| 3. No Fault | 0.1% |

FINAL DIAGNOSIS

STATUS : FAULT DETECTED
FAULT : Fuel System Malfunction
CONFIDENCE : 96.3%
ALERT : HIGH [RED]
REASON : Abnormal fuel trim and O2 readings indicate fuel mixture problems
ACTION : Reduce speed. Schedule fuel injector or O2 sensor inspection.

ANOMALOUS PARAMETERS:

- Engine Coolant Temperature: 20000.00 C

MAIN MENU

1. Manual diagnosis (enter sensor values yourself)
2. Random simulation (system randomizes parameters)
3. Exit

Select option (1/2/3): 1

Enter current OBD-II sensor readings:
(Press ENTER to use default normal value)

Engine Coolant Temperature (C) [normal: 92.0]: 92
Engine RPM (rpm) [normal: 1500.0]:
Engine Load (%) [normal: 35.0]: 1
Intake Air Temperature (C) [normal: 25.0]: 1
Short Term Fuel Trim (%) [normal: 0.0]: 1
Fuel Tank Pressure (kPa) [normal: 0.0]: 11111
Mass Air Flow Rate (g/s) [normal: 8.0]:
Throttle Position (%) [normal: 20.0]:
Battery Voltage (V) [normal: 14.0]:
O2 Sensor Voltage (V) [normal: 0.5]:
Vehicle Speed (km/h) [normal: 60.0]:
Component Wear Index (min) [normal: 50.0]: 1

DIAGNOSIS REPORT – MANUAL INPUT

LEVEL 1: Parameter Threshold Analysis (SAE J1979)

[HIGH [RED]] Fuel Tank Pressure
Value: 11111.00 kPa far outside normal [-1.0, 1.0]

LEVEL 2: Neural Network Fault Prediction (MLP)

Top predicted faults:

- | | |
|------------------------------|-------|
| 1. Fuel System Malfunction | 99.1% |
| 2. Exhaust / Emissions Fault | 0.5% |
| 3. No Fault | 0.3% |

FINAL DIAGNOSIS

STATUS : FAULT DETECTED
FAULT : Fuel System Malfunction
CONFIDENCE : 99.1%
ALERT : HIGH [RED]
REASON : Abnormal fuel trim and O2 readings indicate fuel mixture problems
ACTION : Reduce speed. Schedule fuel injector or O2 sensor inspection.

ANOMALOUS PARAMETERS:

- Fuel Tank Pressure: 11111.00 kPa

MAIN MENU

1. Manual diagnosis (enter sensor values yourself)
 2. Random simulation (system randomizes parameters)
 3. Exit
-

Select option (1/2/3): 1

Enter current OBD-II sensor readings:
(Press ENTER to use default normal value)

Engine Coolant Temperature (C) [normal: 92.0]:
Engine RPM (rpm) [normal: 1500.0]:
Engine Load (%) [normal: 35.0]:
Intake Air Temperature (C) [normal: 25.0]: 100
Short Term Fuel Trim (%) [normal: 0.0]:
Fuel Tank Pressure (kPa) [normal: 0.0]:
Mass Air Flow Rate (g/s) [normal: 8.0]:
Throttle Position (%) [normal: 20.0]:
Battery Voltage (V) [normal: 14.0]:
O2 Sensor Voltage (V) [normal: 0.5]:
Vehicle Speed (km/h) [normal: 60.0]:
Component Wear Index (min) [normal: 50.0]:

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DIAGNOSIS REPORT – MANUAL INPUT

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LEVEL 1: Parameter Threshold Analysis (SAE J1979)

[HIGH [RED]] Intake Air Temperature
Value: 100.00 C >> RED threshold (80.0)

LEVEL 2: Neural Network Fault Prediction (MLP)

Top predicted faults:

- | | |
|-----------------------------|-------|
| 1. Air Intake Fault | 99.8% |
| 2. No Fault | 0.1% |
| 3. Engine Mechanical Stress | 0.0% |

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FINAL DIAGNOSIS

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STATUS : FAULT DETECTED
FAULT : Air Intake Fault
CONFIDENCE : 99.8%
ALERT : HIGH [RED]
REASON : Abnormal airflow and intake temperature suggest air system restriction
ACTION : Check air filter and throttle body. Inspect MAF sensor.

ANOMALOUS PARAMETERS:

- Intake Air Temperature: 100.00 C

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MAIN MENU

1. Manual diagnosis (enter sensor values yourself)
2. Random simulation (system randomizes parameters)
3. Exit

Select option (1/2/3): 1

Enter current OBD-II sensor readings:
(Press ENTER to use default normal value)

Engine Coolant Temperature (C) [normal: 92.0]:
Engine RPM (rpm) [normal: 1500.0]:
Engine Load (%) [normal: 35.0]:
Intake Air Temperature (C) [normal: 25.0]:
Short Term Fuel Trim (%) [normal: 0.0]:
Fuel Tank Pressure (kPa) [normal: 0.0]:
Mass Air Flow Rate (g/s) [normal: 8.0]:
Throttle Position (%) [normal: 20.0]:
Battery Voltage (V) [normal: 14.0]:
O2 Sensor Voltage (V) [normal: 0.5]:
Vehicle Speed (km/h) [normal: 60.0]:
Component Wear Index (min) [normal: 50.0]:

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DIAGNOSIS REPORT – MANUAL INPUT

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LEVEL 1: Parameter Threshold Analysis (SAE J1979)

All parameters within normal operating ranges.

LEVEL 2: Neural Network Fault Prediction (MLP)

Top predicted faults:

- | | |
|--------------------------------|-------|
| 1. Electrical / Charging Fault | 70.7% |
| 2. No Fault | 9.8% |
| 3. Engine Overheating | 9.1% |

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FINAL DIAGNOSIS

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STATUS : NORMAL
FAULT : No fault detected
ALERT : NONE
ACTION : No action required. Continue monitoring.

[INFO] Neural network supplementary output:

Top prediction: Electrical / Charging Fault (70.7%)
Note: Ignored – no threshold violations detected.

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MAIN MENU

1. Manual diagnosis (enter sensor values yourself)
 2. Random simulation (system randomizes parameters)
 3. Exit
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Select option (1/2/3): 1

Enter current OBD-II sensor readings:
(Press ENTER to use default normal value)

Engine Coolant Temperature (C) [normal: 92.0]: 1
Engine RPM (rpm) [normal: 1500.0]: 1
Engine Load (%) [normal: 35.0]: 1
Intake Air Temperature (C) [normal: 25.0]: 1
Short Term Fuel Trim (%) [normal: 0.0]: 2
Fuel Tank Pressure (kPa) [normal: 0.0]: 2
Mass Air Flow Rate (g/s) [normal: 8.0]:
Throttle Position (%) [normal: 20.0]:
Battery Voltage (V) [normal: 14.0]:
O2 Sensor Voltage (V) [normal: 0.5]:
Vehicle Speed (km/h) [normal: 60.0]: 1
Component Wear Index (min) [normal: 50.0]: 1

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DIAGNOSIS REPORT – MANUAL INPUT

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LEVEL 1: Parameter Threshold Analysis (SAE J1979)

[MEDIUM [ORANGE]] Fuel Tank Pressure
Value: 2.00 kPa outside normal range

LEVEL 2: Neural Network Fault Prediction (MLP)

Top predicted faults:

1. Fuel System Malfunction	55.6%
2. Exhaust / Emissions Fault	31.6%
3. No Fault	12.1%

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FINAL DIAGNOSIS

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STATUS : ANOMALY DETECTED (fault type unclear)
ALERT : MEDIUM [ORANGE]
CONFIDENCE : 55.6% (below threshold for fault ID)
ACTION : Inspect anomalous parameters. Seek professional diagnosis.

ANOMALOUS PARAMETERS:

- Fuel Tank Pressure: 2.00 kPa

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MAIN MENU

1. Manual diagnosis (enter sensor values yourself)
2. Random simulation (system randomizes parameters)
3. Exit

Select option (1/2/3): 3

System closed. Goodbye.

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